Champion-Ensemble Initialization with Transactional Probability-Map Scribble Editing for Interactive Whole-Body PET/CT Lesion Segmentation

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Code: https://github.com/il-miscusi/autopetv-algorithm

Abstract. We describe a submission to the autoPET V challenge on interactive, scribble-guided lesion segmentation in whole-body PET/CT. The initial prediction is the publicly released five-fold autoPET-III winning ensemble (LesionTracer, nnU-Net ResEncL) [1], routed per case to an FDG- or PSMA-specific connected-component policy by a fixed three-classifier tracer vote. Corrective scribbles are then applied as *transactions* against the previous accepted mask, which is persisted through the platform's output directory: each scribble locally modulates the segmentation threshold of the frozen ensemble probability map inside a sign-specific Gaussian support (TACE, adapted from public Apache-2.0 work by Chen [2]), and a transaction gate rejects removals that split an existing lesion component, additions that merge distinct lesions, and any proposal that creates a new enclosed hole, falling back to the previous mask. These guards target the challenge's equally weighted lesion-detection metric (AUC-DMM) rather than Dice alone. The method requires no new training; all weights are public and hash-pinned, and the container performs no network access at inference.

Keywords: Interactive segmentation · PET/CT · Lesion segmentation · Scribbles.

1 Introduction

autoPET V evaluates algorithms that produce an initial whole-body PET/CT lesion segmentation and refine it over five corrective steps. Each step supplies one additional simulated or clinician-drawn scribble; final ranking combines the per-case area under the Dice trajectory (AUC-Dice) and under a lesion-level detection-matching trajectory (AUC-DMM) with equal weight. Two properties of this protocol shaped our design. First, the initial prediction contributes to every point of both trajectories, so initialization quality dominates the integral: we therefore initialize with the strongest publicly available model rather than the challenge baseline. Second, edits that improve voxel overlap can still destroy detection score — deleting a whole predicted component removes a detected lesion, and growing a region until it touches a neighbour merges two — so every interactive edit is checked at the lesion-component level before it is accepted.

2 Method

2.1 Initialization

The zero-scribble invocation runs the five-fold autoPET-III winning ensemble (LesionTracer; nnU-Net ResEncL trained on multi-tracer data with organ supervision) [1], using the authors' public Zenodo checkpoint, in an isolated subprocess. A fixed majority vote of three classical classifiers (histogram gradient boosting, linear SVM, random forest) over 99 whole-body intensity-distribution features routes the case to a component policy that keeps 18-connected components of at least 25 voxels. The router and component floor are adapted from public Apache-2.0 work [2].

2.2 Transactional scribble editing

The previous accepted mask is reloaded from the platform-persisted output directory, with geometry verification and a fail-closed fallback to the initial ensemble mask. Cumulative tumor/background scribble voxels are converted to disjoint positive/negative cores. Each core locally raises or lowers the binarization threshold of the frozen ensemble probability map inside a Gaussian support around the scribble (add: $\sigma = 12\,\text{mm}$ with a $3\,\text{mm}$ plateau; remove: $\sigma = 8\,\text{mm}$), so edits are probability-aware and spatially bounded rather than fixed-radius morphology. The resulting proposal is applied to the previous mask as a single transaction under four checks: (i) when a background-scribble proposal removes a strict majority of its prompted component, the remainder is removed as well to resolve the detection-level false positive; (ii) other removals may not split a previously connected component (the offending removal is rolled back); (iii) additions may not merge two previously distinct components; (iv) if the accepted candidate would create a new enclosed hole, the transaction falls back to the previous mask. Scribble voxels themselves are always enforced (tumor = 1, background = 0).

3 Validation

The preliminary test-set leaderboard evaluates a single non-interactive iteration. At the time of writing our initialization reproduces the expected champion-ensemble operating point on that leaderboard, which public entries place at approximately 0.85 Dice; the interactive editing path is exercised only in the organizers' offline six-step evaluation. We replayed that protocol with the organizers' published simulation harness (six steps, all three scribble strategies) on four held-out labeled whole-body PET/CT cases (two FDG, two PSMA) on a

T4 GPU. Across all twelve case–strategy pairs, Dice improved from iteration 0 to iteration 5 in every pair and the lesion-level f_1 never decreased (worst case flat); mean trapezoid AUC-Dice was 0.734 against 0.705 for a no-edit control that holds the initial mask fixed, and mean AUC-f_1 was 0.628 against 0.559. Per-iteration edit cost was 1–4 s with a peak of 12.4 GB GPU memory, within the platform's T4 and 15-minute-per-case limits. We make no monotonicity claim: the transaction gate bounds the damage a single edit can do to the detection metric, but does not guarantee per-step improvement, and clinician scribbles — collected in advance against baseline predictions and replayed unchanged — need not target our model's largest current error at all. The gate's fail-closed fallbacks are designed to degrade to the previous accepted mask in exactly those cases.

4 Discussion

Our earlier heuristic editor (PET-adaptive region growing plus component-removal rules on the challenge baseline) validated worse than the stateless baseline in our own six-step replay and is retained in the repository only as an ablation. The submitted method replaces it with the combination above; the design lesson we take from that failure is that interactive edits must be measured under the exact evaluation protocol (step count, scribble regimes, and both metrics) before any claim is made about them.

Acknowledgements. We build directly on the public autoPET-III winning model of the MIC-DKFZ team [1] (Zenodo 14007247, CC-BY-4.0) and on the public Apache-2.0 autoPET-V scaffolding, tracer router, and TACE editor of Yixin Chen [2], whose trained LocalEdit specialist checkpoints we deliberately do not use. We thank the autoPET V organizers for the interactive evaluation harness.

References

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